

732A96/TDDE15 Advanced Machine Learning

Graphical Models

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Lecture 1: Bayesian Networks

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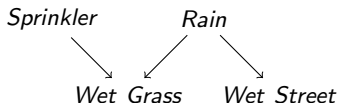
- Causal Structures
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 - Definition
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Literature

- ▶ Main source
 - ▶ Bishop, C. M. *Pattern Recognition and Machine Learning*. Springer, 2006. Chapter 8.
- ▶ Additional source
 - ▶ Koski, T. J. T. and Noble, J. M. A Review of Bayesian Networks and Structure Learning. *Mathematica Applicanda* 40, 51-103, 2012.

Causal Structures

- ▶ Assume that we want to represent the causal relations between a set of random variables, e.g. the variables may represent the state of the components of a system.
- ▶ A natural and intuitive representation consists of a **graph** where the nodes are the random variables, and the edges are the causal relations between the variables. We call such a graph a **causal structure**.



- ▶ **Exercise.** Produce a causal structure for the domain *Temperature*, *Ice cream sales* and *Soda sales*.
- ▶ **Exercise.** Produce a causal structure for Boyle's law, which relates the pressure and volume of a gas as $Pressure \cdot Volume = constant$ if the temperature and amount of gas remain unchanged within a closed system.

Bayesian Networks: Definition

DAG	Parameter values for the conditional probability distributions
<pre> graph TD Sprinkler --> WetGrass[Wet Grass] Rain --> WetGrass Rain --> WetStreet[Wet Street] </pre>	$q(s) = (0.3, 0.7) = (\theta_{s_0}, \theta_{s_1})$ $q(r) = (0.5, 0.5) = (\theta_{r_0}, \theta_{r_1})$ $q(wg r_0, s_0) = (0.1, 0.9) = (\theta_{wg_0 r_0, s_0}, \theta_{wg_1 r_0, s_0})$ $q(wg r_0, s_1) = (0.7, 0.3) = (\theta_{wg_0 r_0, s_1}, \theta_{wg_1 r_0, s_1})$ $q(wg r_1, s_0) = (0.8, 0.2) = (\theta_{wg_0 r_1, s_0}, \theta_{wg_1 r_1, s_0})$ $q(wg r_1, s_1) = (0.9, 0.1) = (\theta_{wg_0 r_1, s_1}, \theta_{wg_1 r_1, s_1})$ $q(ws r_0) = (0.1, 0.9) = (\theta_{ws_0 r_0}, \theta_{ws_1 r_0})$ $q(ws r_1) = (0.7, 0.3) = (\theta_{ws_0 r_1}, \theta_{ws_1 r_1})$ $p(s, r, wg, ws) = q(s)q(r)q(wg s, r)q(ws r)$

- ▶ A **Bayesian network (BN)** over a finite set of **discrete** random variables $X = X_{1:n} = \{X_1, \dots, X_n\}$ consists of
 - ▶ a directed acyclic graph (DAG) G whose nodes are the elements in X , and
 - ▶ parameter values θ specifying probability distributions $q(x_i|pa_i)$, where Pa_i are the parents of X_i in G , i.e. the nodes with an edge into X_i .
- ▶ The BN represents a **causal** model of the system.
- ▶ And also a **probabilistic** model of the system as $p(x) = \prod_i q(x_i|pa_i)$.

Bayesian Networks: Definition

- ▶ We now show that $p(x) = \prod_i q(x_i|pa_i)$ is a probability distribution.
- ▶ Clearly, $0 \leq \prod_i q(x_i|pa_i) \leq 1$.
- ▶ Assume without loss of generality that $Pa_i \subseteq X_{1:i-1}$ for all i . Then

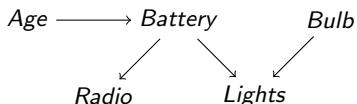
$$\sum_x \prod_i q(x_i|pa_i) = \sum_{x_1} [q(x_1) \dots \sum_{x_{n-1}} [q(x_{n-1}|pa_{n-1}) \sum_{x_n} q(x_n|pa_n)] \dots] = 1$$

- ▶ Moreover, $p(x_j|pa_j) = q(x_j|pa_j)$. To see it, note that

$$\begin{aligned} p(x_j|pa_j) &= \frac{p(x_j, pa_j)}{p(pa_j)} = \frac{\sum_{x \setminus \{x_j, pa_j\}} \prod_i q(x_i|pa_i)}{\sum_{x \setminus pa_j} \prod_i q(x_i|pa_i)} \\ &= \frac{\sum_{x_{1:j} \setminus \{x_j, pa_j\}} \prod_{i \leq j} q(x_i|pa_i)}{\sum_{x_{1:j} \setminus pa_j} \prod_{i \leq j} q(x_i|pa_i)} = q(x_j|pa_j) \end{aligned}$$

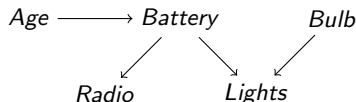
Bayesian Networks: Separation

- ▶ We now show that many of the independencies in p can be read off G without numerical calculations. Consider the following DAG.



- ▶ **Chain:** $Age \rightarrow Battery \rightarrow Radio$
 - ▶ $Age \not\perp Radio | \emptyset$
 - ▶ $Age \perp Radio | Battery$
- ▶ **Fork:** $Radio \leftarrow Battery \rightarrow Lights$
 - ▶ $Radio \not\perp Lights | \emptyset$
 - ▶ $Radio \perp Lights | Battery$
- ▶ **Collider:** $Battery \rightarrow Lights \leftarrow Bulb$
 - ▶ $Battery \perp Bulb | \emptyset$
 - ▶ $Battery \not\perp Bulb | Lights$
- ▶ **Chain + collider:** $Age \rightarrow Battery \rightarrow Lights \leftarrow Bulb$
 - ▶ $Age \perp Bulb | \emptyset$
 - ▶ $Age \not\perp Bulb | Lights$
 - ▶ $Age \perp Bulb | Lights, Battery$

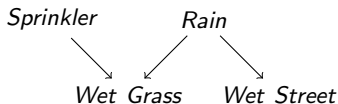
Bayesian Networks: Separation



- ▶ A path in G is a sequence of distinct and adjacent nodes, i.e. the direction of the edge is irrelevant. A node B is a descendant of a node A in G if there is a path $A \rightarrow \dots \rightarrow B$.
 - ▶ E.g., $Age \rightarrow Battery \rightarrow Lights \leftarrow Bulb$ is a path.
 - ▶ E.g., $Lights$ is a descendant of Age .
- ▶ Let ρ be a path in G between the nodes α and β .
- ▶ A node B in ρ is a **collider** when $A \rightarrow B \leftarrow C$ is a subpath of ρ .
 - ▶ E.g., $Lights$ is a collider in the path $Age \rightarrow Battery \rightarrow Lights \leftarrow Bulb$.
- ▶ Moreover, ρ is **Z-open** with $Z \subseteq X \setminus \{\alpha, \beta\}$ when
 - ▶ no non-collider in ρ is in Z , and
 - ▶ every collider in ρ is in Z or has a descendant in Z .
 - ▶ E.g., the path $Age \rightarrow Battery \rightarrow Lights \leftarrow Bulb$ is Z-open with $Z = \{Lights\}$.
- ▶ Let U , V and Z be three disjoint subsets of X . Then, U and V are **separated** given Z in G (i.e. $U \perp_G V | Z$) when there is no Z-open path in G between a node in U and a node in V .
 - ▶ E.g., $Age, Battery \perp_G Bulb | \emptyset$.

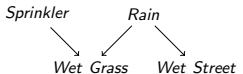
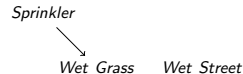
Bayesian Networks: Separation

- ▶ The separation criterion is **sound**, i.e. if $U \perp_G V | Z$ then $U \perp_p V | Z$.
- ▶ For instance, $S \perp_p R$ and $S \perp_p WS | WG, R$ follow from the DAG



- ▶ Note that we read independencies from G , never dependencies. That is, $S \not\perp_G R | WG$ and $S \not\perp_G WS | WG$ follow from the DAG, but not $S \not\perp_p R | WG$ or $S \not\perp_p WS | WG$.
- ▶ **Exercise.** Prove that $A \perp_p B | C$ for the DAGs $A \rightarrow C \rightarrow B$, $A \leftarrow C \rightarrow B$ and $A \leftarrow C \leftarrow B$, i.e. prove that $p(a, b | c) = p(a | c)p(b | c)$.
- ▶ **Exercise.** Prove that $A \perp_p B | \emptyset$ for the DAG $A \rightarrow C \leftarrow B$, i.e. prove that $p(a, b) = p(a)p(b)$.
- ▶ **Exercise.** Find the minimal set of nodes that separates a given node from the rest. This set is called the Markov blanket of the given node.
- ▶ **Exercise.** How many free parameters do we have in the wet grass BN ? How many do we have if we specify the distribution without the assistance of a BN, i.e. as a table ?

Bayesian Networks: Causal Inference

Original	After $do(r_1)$
 $q(s) = (0.3, 0.7)$ $q(r) = (0.5, 0.5)$ $q(wg r_0, s_0) = (0.1, 0.9)$ $q(wg r_0, s_1) = (0.7, 0.3)$ $q(wg r_1, s_0) = (0.8, 0.2)$ $q(wg r_1, s_1) = (0.9, 0.1)$ $q(ws r_0) = (0.1, 0.9)$ $q(ws r_1) = (0.7, 0.3)$ $p(s, r, wg, ws) = q(s)q(r)q(wg s, r)q(ws r)$	 $q(s) = (0.3, 0.7)$ $q(wg s_0) = (0.8, 0.2)$ $q(wg s_1) = (0.9, 0.1)$ $q(ws) = (0.7, 0.3)$ $p(s, wg, ws) = q(s)q(wg s)q(ws)$

- What would be the state of the system if a random variable X_j is **forced** to take the state x_j ?
 - Remove X_j and all the edges from and to X_j from G .
 - Remove $q(x_j|pa_j)$.
 - If $X_j \in Pa_i$, then replace $q(x_i|pa_i)$ with $q(x_i|pa_i \setminus x_j, x_j)$
 - Set $p(x \setminus x_j|do(x_j)) = \prod_i q(x_i|pa_i)$.
- So, the result of $do(x)$ on a BN is a **BN**. No more on causality in this course.

Bayesian Networks: Probabilistic Inference

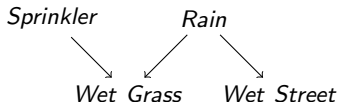
- What is the state of a random variable X_k if a random variable X_i is **observed** to be in the state x_i ?

$$p(x_k|x_i) = \frac{p(x_k, x_i)}{p(x_i)} = \frac{\sum_{x \setminus \{x_i, x_k\}} p(x)}{\sum_{x \setminus x_i} p(x)}$$

- For instance,

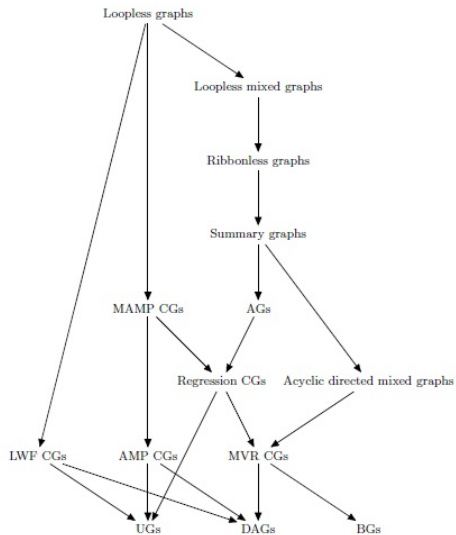
$$p(ws|s) = \frac{\sum_{r, wg} p(r, wg, ws, s)}{\sum_{r, wg, ws} p(r, wg, ws, s)} = \frac{\sum_{r, wg} q(s)q(r)q(wg|s, r)q(ws|r)}{\sum_{r, wg, ws} q(s)q(r)q(wg|s, r)q(ws|r)}$$

for the DAG



- Answering questions like the one above can be computationally hard.
- A BN is an efficient (because it uses the independences encoded) formalism to compute a posterior probability distribution from a prior probability distribution in the light of observations, hence the name. More on probabilistic inference in Lecture 2.

Families of Graphical Models



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Thank you